

# Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

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Math Camp 2026

# How to use these slides

**Main deck:** definitions, recipes, examples, and practice.

**Appendix and proofs** (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

**Not required for Math Camp.** Focus on the main slides and exercises unless you want the extra detail.

## 3.1 Fixed points — equilibrium as a solution

**Fixed point.** A point  $x^*$  is a **fixed point** of a map  $T$  if  $T(x^*) = x^*$ .

**Economics reading.** Many equilibria are fixed-point problems:

- **Market clearing:** a price  $p^*$  with excess demand  $E(p^*) = 0$ .
- **Game theory:** a strategy profile that is a best response to itself (Nash).

**Two useful existence tools (today).**

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

## 3.1 Contraction Mapping Theorem

**Contraction.** A map  $T$  **shrinks distances** if there is  $k \in (0, 1)$  such that  $|T(x) - T(y)| \leq k|x - y|$  for all  $x, y$  in the domain.

**Contraction Mapping Theorem.** If  $T: [a, b] \rightarrow [a, b]$  is a contraction, then

- **Existence:** there is a fixed point  $x^*$  with  $T(x^*) = x^*$ .
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value  $x_0 \in [a, b]$ , the sequence  $x_n = T(x_{n-1})$  converges to  $x^*$ .

## 3.1 Why iteration converges

**Step 1 — Start anywhere.** Pick any  $x_0$ . Define  $x_n = T(x_{n-1})$  for  $n = 1, 2, \dots$

**Step 2 — Jumps shrink exponentially.** Each new step is at most  $k$  times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

Since  $k < 1$ , the moves get **exponentially smaller** — the path barely changes after a while.

**Step 3 — The sequence converges.** Step 2 implies  $x_n \rightarrow x^*$  for some limit  $x^*$  (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives  $x_n$  toward the same  $x^*$ .

**Step 4 — The limit is a fixed point.**  $T$  is continuous, so taking limits in  $x_n = T(x_{n-1})$  gives  $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$ .

## 3.1 Walrasian price adjustment

**Market clearing.** A price  $p^*$  **clears the market** when excess demand is zero:  $E(p^*) = 0$ . Here  $E(p)$  can take **any form** (from supply and demand behind the scenes).

**Price adjustment (nonlinear in  $E$ ).** Update the price each period:

$$p_{t+1} = T(p_t) = p_t + \lambda(1 - e^{-\alpha E(p_t)}), \quad \lambda = \frac{1}{5}, \alpha = 0.1.$$

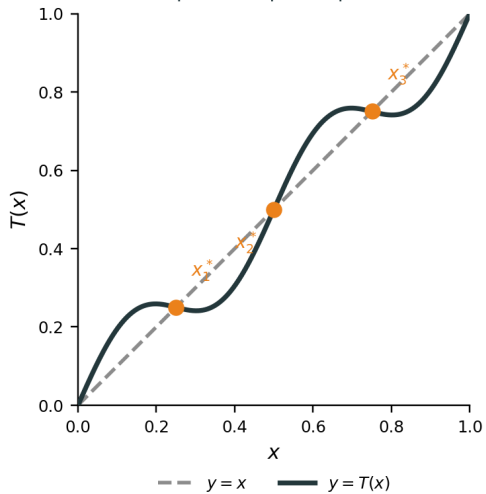
When  $|E|$  is small,  $1 - e^{-\alpha E} \approx \alpha E$  — move a fraction of excess demand toward balance. Clearing satisfies  $E(p^*) = 0 \Leftrightarrow T(p^*) = p^*$ . If  $T$  is a **contraction** on a price range, then  $p_t \rightarrow p^*$  from any starting  $p_0$ .

## 3.1 Brouwer — one dimension

**Brouwer (one dimension).** If  $T: [a, b] \rightarrow [a, b]$  is **continuous**, then  $T$  has **at least one** fixed point  $x^*$  with  $T(x^*) = x^*$ .

**Picture.** Plot  $y = T(x)$  against  $y = x$ . Any crossing is a fixed point. Continuity forces the graph to meet the diagonal at least once on  $[a, b]$ .

Multiple fixed points possible





## 3.1 Brouwer — general statement

**Brouwer (several variables).** Let  $K \subset \mathbb{R}^n$  be **nonempty**, **compact**, and **convex**. If  $T: K \rightarrow K$  is **continuous**, then  $T$  has at least one fixed point  $x^* \in K$  with  $T(x^*) = x^*$ .

**Camp vocabulary.**

- **Compact** in  $\mathbb{R}^n$ : closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in  $K$  stays in  $K$ .

**One dimension.**  $K = [a, b]$  is compact and convex, so the 1D case is a special case.

## 3.1 Brouwer — Cournot duopoly

**Two firms.** Firm  $i$  chooses  $q_i \in [0, \bar{q}]$ . **Nonlinear** inverse demand  $p = a - (q_1 + q_2)^2$  (e.g.  $a = 8$ ,  $\bar{q} = 2$ ).

**Profit and FOC.**  $\pi_i = (a - (q_1 + q_2)^2)q_i$ . Firm 1's best response solves

$$a - 3q_1^2 - 4q_1q_2 - q_2^2 = 0 \quad (\text{similarly for firm 2 with } q_1, q_2 \text{ swapped}).$$

The map  $q_j \mapsto BR_i(q_j)$  is **nonlinear** — no formula like “ $(a - c - q_j)/2$ ”.

**Fixed point = Nash.**  $T(q_1, q_2) = (BR_1(q_2), BR_2(q_1))$  is continuous on  $K = [0, \bar{q}]^2$ . **Brouwer** guarantees an equilibrium, but you generally solve  $T(q_1^*, q_2^*) = (q_1^*, q_2^*)$  by **algebra or numerics** (here one symmetric solution is  $q_1^* = q_2^* = \sqrt{a/8}$ ).

## 3.2 Separating Hyperplane Theorem

**Setup.** Nonempty **convex** sets  $A, B \subset \mathbb{R}^n$  with  $A \cap B = \emptyset$ .

**Theorem.** If at least one of  $A, B$  is **compact**, then there exist  $\mathbf{p} \neq \mathbf{0}$  and  $\beta \in \mathbb{R}$  such that

$$\mathbf{p} \cdot \mathbf{a} \leq \beta \leq \mathbf{p} \cdot \mathbf{b} \quad \text{for all } \mathbf{a} \in A, \mathbf{b} \in B.$$

Geometrically, the hyperplane  $\mathbf{p} \cdot \mathbf{x} = \beta$  separates  $A$  from  $B$ . (If only  $B$  is compact, swap the labels of  $A$  and  $B$ .)

Proof and strict separation: [appendix \(for interest only\)](#).

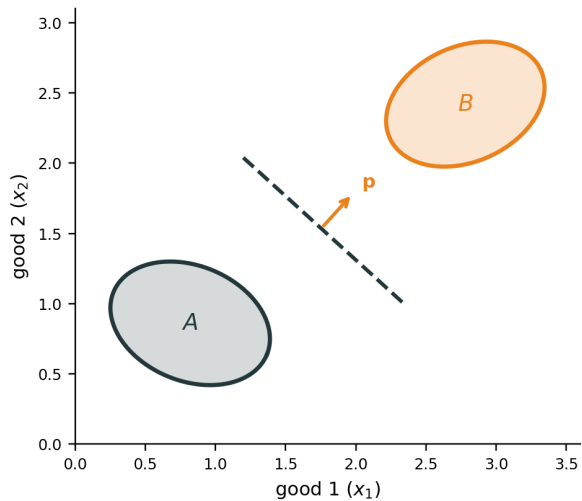
## 3.2 Separating Hyperplane Theorem — two goods

With  $x = (x_1, x_2)$ , separation is a line  $p_1x_1 + p_2x_2 = \beta$ : all of  $A$  on one side, all of  $B$  on the other.

**Two useful special cases.**

- **Single point.**  $B = \{b_0\}$  is compact. If  $b_0 \notin A$  and  $A$  is closed and convex, a separator exists even when  $A$  is not compact.
- **Strict separation.** If the sets are a positive distance apart, the same  $\mathbf{p}$  can separate **strictly**:  
 $\mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b$ .

Disjoint convex sets separated by a line



## 3.2 Example — producer theory (numerical illustration)

**Netput plan**  $y = (y_1, y_2)$ : input  $y_1 \leq 0$ , output  $y_2 \geq 0$  (second quadrant).

**Technology.** With  $L = -y_1$  units of labor, max output is  $f(L) = 2\sqrt{L}$ , so

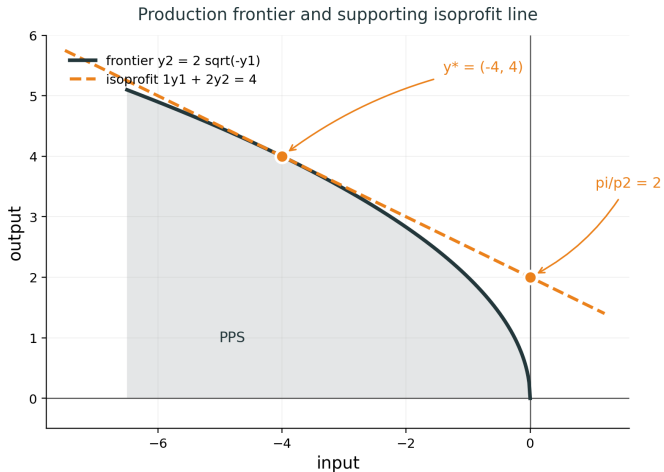
$$Y = \{(y_1, y_2) : y_1 \leq 0, 0 \leq y_2 \leq 2\sqrt{-y_1}\} \quad (\text{free disposal on output}).$$

**Prices.** Wage  $p_1 = 1$ , output price  $p_2 = 2$ . Profit  $\pi = p_1 y_1 + p_2 y_2$  (since  $y_1 < 0$ , the term  $p_1 y_1$  is labor cost).

**Efficient plan.** Use  $L = 4$ :  $y^* = (-4, 4)$ . Then  $\pi = 1 \cdot (-4) + 2 \cdot 4 = 4$ .

**Isoprofit line.**  $y_1 + 2y_2 = 4$ , or  $y_2 = 2 - \frac{1}{2}y_1$ . At  $y_1 = 0$ :  $y_2$ -axis intercept  $y_2 = \pi/p_2 = 2$ . Slope  $-p_1/p_2 = -\frac{1}{2}$  matches the frontier at  $y^*$  (supporting prices).

## 3.2 Example — producer theory (graph)



## 3.2 Example — producer theory

**Setup.** A firm's **production possibility set**  $Y \subset \mathbb{R}^n$  is **convex** (and closed in full treatments).

**Statement.** If  $Y$  is convex, the theorem guarantees that every **technically efficient** production plan  $y \in Y$  is also a **profit-maximizing** choice for some nonzero price vector  $\mathbf{p}$ :

$$\mathbf{p} \cdot y \geq \mathbf{p} \cdot y' \quad \text{for all } y' \in Y.$$

**Separation picture.** Let  $B$  be outputs **strictly better than**  $y$  (more of some good, no less of any). If  $y$  is efficient,  $Y \cap B = \emptyset$ . A separating hyperplane gives supporting prices  $\mathbf{p}$  at  $y$ .

Proof: [appendix](#) (for interest only).



# Linear Algebra Basics

### 3.3 Vectors in $\mathbb{R}^n$

**Vector.** An ordered list of  $n$  real numbers. We write a **column vector** as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Its **transpose**  $\mathbf{x}^T = (x_1, \dots, x_n)$  is a **row vector**.

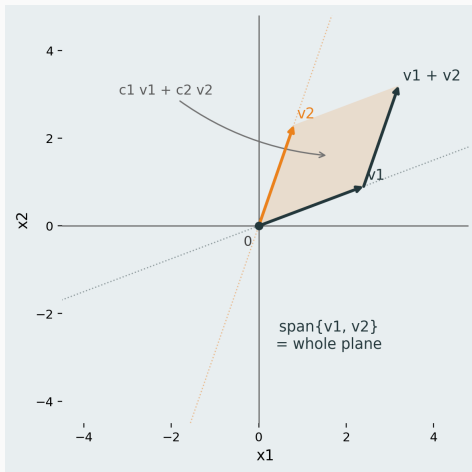
**Linear combination.** Given vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$  and scalars  $c_1, \dots, c_k$ ,

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$

is a **linear combination** of the  $\mathbf{v}_j$ .

**Span.** All linear combinations of  $\mathbf{v}_1, \dots, \mathbf{v}_k$  form the **span** of those vectors.

### 3.3 Span in the plane (two vectors)



### 3.3 Matrix–vector product — columns of $A$

**Setup.**  $A \in \mathbb{R}^{m \times n}$  has columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^m$  and  $\mathbf{x} \in \mathbb{R}^n$  is a column vector.

**Column view.**  $A\mathbf{x}$  is a linear combination of the **columns** of  $A$ :

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n \in \mathbb{R}^m.$$

The weights are the entries of  $\mathbf{x}$ .

**Example.**  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$ ,  $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ :

$$A\mathbf{x} = x_1 \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

### 3.3 Row vector times matrix — rows of $A$

**Setup.**  $A \in \mathbb{R}^{m \times n}$ ,  $\mathbf{x} \in \mathbb{R}^n$  (column), so  $A\mathbf{x} \in \mathbb{R}^m$ .

**Row view (transpose both sides).**  $(A\mathbf{x})^T = \mathbf{x}^T A^T$ :

$$\mathbf{x}^T A^T = x_1 \mathbf{a}_1^T + x_2 \mathbf{a}_2^T + \cdots + x_n \mathbf{a}_n^T \in \mathbb{R}^{1 \times m}.$$

Dimensions:  $(1 \times n)(n \times m) = (1 \times m)$ . **Note:**  $\mathbf{x}^T A$  does **not** match when  $A \in \mathbb{R}^{m \times n}$  with  $m \neq n$ .

**Example.** From the previous slide,

$$\mathbf{x}^T = (x_1, x_2), \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

Then

$$\mathbf{x}^T A^T = x_1(1, 3, 5) + x_2(2, 4, 6).$$

## 3.3 Two views of the same product

**Same numbers, two interpretations.** If  $A \in \mathbb{R}^{m \times n}$  and  $\mathbf{x} \in \mathbb{R}^n$ :

- **Column view:**  $A\mathbf{x} = \sum_{j=1}^n x_j \mathbf{a}_j \in \mathbb{R}^m$ .
- **Entry view:**  $(A\mathbf{x})_i = \mathbf{r}_i^T A\mathbf{x} = \sum_{j=1}^n a_{ij}x_j$  (dot product of row  $i$  of  $A$  with  $\mathbf{x}$ ), where  $\mathbf{r}_i \in \mathbb{R}^m$  has  $i$ -th entry 1 and all other entries 0 ( $\mathbf{r}_i = \mathbf{e}_i$ ).

**Transpose link.**  $(A\mathbf{x})^T = \mathbf{x}^T A^T$ : a **row vector** on the left combines **rows of  $A^T$**  (i.e. transposes of columns of  $A$ ).

### 3.3 Matrix multiplication $AB$ — example

**Example.**  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ ,  $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$ .

**Column view (both columns of  $AB$ ).**

$$(AB)\mathbf{e}_1 = A \begin{pmatrix} 5 \\ 7 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 7 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 19 \\ 43 \end{pmatrix},$$

$$(AB)\mathbf{e}_2 = A \begin{pmatrix} 6 \\ 8 \end{pmatrix} = 6 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 8 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 22 \\ 50 \end{pmatrix}.$$

So  $AB = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$ .

**Row view (practice).** Find rows 1 and 2 of  $AB$  using  $(1, 2)B$  and  $(3, 4)B$ .

**Note (skip in class).**  $(1, 2)B = (19, 22)$  and  $(3, 4)B = (43, 50)$  — same matrix as above.

### 3.3 Matrix multiplication $AB$

**Setup.**  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{n \times p}$ , so  $AB \in \mathbb{R}^{m \times p}$ .

**Column view.** Column  $j$  of  $AB$  is  $A$  times column  $j$  of  $B$ :

$$(AB)\mathbf{e}_j = A(B\mathbf{e}_j) = A\mathbf{b}_j = b_{1j}\mathbf{a}_1 + b_{2j}\mathbf{a}_2 + \cdots + b_{nj}\mathbf{a}_n.$$

Each column of  $AB$  is a linear combination of **columns of  $A$**  (weights from column  $j$  of  $B$ ).

**Row view.** Row  $i$  of  $AB$  is row  $i$  of  $A$  times  $B$ :

$$\mathbf{r}_i^T(AB) = (\mathbf{r}_i^T A)B = a_{i1}\mathbf{s}_1^T + \cdots + a_{in}\mathbf{s}_n^T,$$

where  $\mathbf{r}_i \in \mathbb{R}^m$  has  $i$ -th entry 1 and all other entries 0 ( $\mathbf{r}_i = \mathbf{e}_i$ ), so  $\mathbf{r}_i^T$  selects row  $i$ ; and  $\mathbf{s}_k^T$  is row  $k$  of  $B$ : a linear combination of **rows of  $B$** .



## 3.3 Basis

**Idea.** A **basis** of  $\mathbb{R}^n$  is a small set of vectors from which every vector in  $\mathbb{R}^n$  can be built as a **linear combination** in exactly one way.

**Standard basis of  $\mathbb{R}^n$ .** For  $j = 1, \dots, n$ , define  $\mathbf{e}_j$  to have a 1 in entry  $j$  and 0 elsewhere:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

**Coordinates.** Every  $\mathbf{x} \in \mathbb{R}^n$  can be written uniquely as

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n,$$

where  $x_1, \dots, x_n$  are the **coordinates** (entries) of  $\mathbf{x}$ .

### 3.3 Basis — example ( $\mathbb{R}^2$ )

**Key point.** A basis is **not unique**, but for a **fixed** basis every vector has a **unique** representation as a linear combination.

**Two bases of  $\mathbb{R}^2$ .** Standard:  $\mathbf{e}_1 = (1, 0)$ ,  $\mathbf{e}_2 = (0, 1)$ . Alternative:  $\mathbf{v}_1 = (1, 0)$ ,  $\mathbf{v}_2 = (1, 1)$ .

**Same vector, two representations.** For  $\mathbf{b} = (2, 3)$ :

$$\mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_2 \quad \text{and} \quad \mathbf{b} = (-1)\mathbf{v}_1 + 3\mathbf{v}_2.$$

Different bases give different weights  $x_1, \dots, x_n$ , but within one basis the weights are **unique**.

**Requirement for a basis  $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ .** Every  $\mathbf{b} \in \mathbb{R}^n$  can be written as  $\mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$  in exactly one way.

## 3.3 Basis — existence and uniqueness

**Requirement for a basis**  $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ . Every  $\mathbf{b} \in \mathbb{R}^n$  can be written as  $\mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$  in exactly one way.

Two parts.

- **Existence:** every  $\mathbf{b} \in \mathbb{R}^n$  can be written as some linear combination of  $\mathbf{a}_1, \dots, \mathbf{a}_n$ .
- **Uniqueness:** there is **only one** choice of weights  $x_1, \dots, x_n$ .

For  $A\mathbf{x} = \mathbf{b}$  with  $A \in \mathbb{R}^{m \times n}$ . Then  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{b} \in \mathbb{R}^m$ , and  $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n]$  with  $\mathbf{a}_j \in \mathbb{R}^m$  gives

$$A\mathbf{x} = \mathbf{b} \iff \mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n.$$

## 3.3 Existence — column space

**Setup.**  $A \in \mathbb{R}^{m \times n}$ ,  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{b} \in \mathbb{R}^m$ ,  $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$  with  $\mathbf{a}_j \in \mathbb{R}^m$ .

**Column space.**

$$\text{Col}(A) = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} \subseteq \mathbb{R}^m.$$

**Existence of a solution.**  $A\mathbf{x} = \mathbf{b}$  has **at least one** solution  $\iff \mathbf{b} \in \text{Col}(A)$ .

**Meaning.**  $\mathbf{b} \in \text{Col}(A)$  means  $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$  for some  $\mathbf{x}$  — the **existence** part of the basis requirement, but only for this one  $\mathbf{b} \in \mathbb{R}^m$  (and only when  $\text{Col}(A) = \mathbb{R}^m$  does it hold for **all**  $\mathbf{b}$ ).

### 3.3 Uniqueness — linearly independent columns

**Setup.**  $A \in \mathbb{R}^{m \times n}$ ,  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{b} \in \mathbb{R}^m$ .

**Linearly independent.** Vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k$  are **linearly independent** if

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0} \implies c_1 = c_2 = \dots = c_k = 0.$$

**Linearly dependent:** not independent — some nontrivial combination equals  $\mathbf{0}$ .

**Uniqueness of a solution.** If  $A\mathbf{x} = \mathbf{b}$  has a solution, it is **unique** iff the columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$  are **linearly independent**.

### 3.3 Linear independence $\Leftrightarrow$ unique solution

**Claim.** Fix  $\mathbf{b} \in \mathbb{R}^m$ . If  $A\mathbf{x} = \mathbf{b}$  has **at least one** solution, then it has a **unique** solution iff  $\mathbf{a}_1, \dots, \mathbf{a}_n$  are linearly independent.

$(\Rightarrow)$  **Independent**  $\Rightarrow$  **unique**. If  $A\mathbf{x} = \mathbf{b}$  and  $A\mathbf{y} = \mathbf{b}$ , then  $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$ , i.e.

$$(x_1 - y_1)\mathbf{a}_1 + \dots + (x_n - y_n)\mathbf{a}_n = \mathbf{0}.$$

Independence gives  $x_j - y_j = 0$  for all  $j$ , so  $\mathbf{x} = \mathbf{y}$ .

$(\Leftarrow)$  **Dependent**  $\Rightarrow$  **not unique**. If the columns are dependent, some  $\mathbf{d} \neq \mathbf{0}$  satisfies  $A\mathbf{d} = \mathbf{0}$ . If  $\mathbf{x}$  solves  $A\mathbf{x} = \mathbf{b}$ , then so does  $\mathbf{x} + \mathbf{d}$ :

$$A(\mathbf{x} + \mathbf{d}) = A\mathbf{x} + A\mathbf{d} = \mathbf{b} + \mathbf{0} = \mathbf{b},$$

and  $\mathbf{x} + \mathbf{d} \neq \mathbf{x}$ . So the solution is **not** unique.

### 3.3 Linear independence — examples ( $\mathbb{R}^2$ )

**Independent (bases).**

- Standard:  $\mathbf{e}_1 = (1, 0)$ ,  $\mathbf{e}_2 = (0, 1)$ .
- Alternative:  $\mathbf{v}_1 = (1, 0)$ ,  $\mathbf{v}_2 = (1, 1)$ .

For each pair,  $c_1\mathbf{w}_1 + c_2\mathbf{w}_2 = \mathbf{0}$  implies  $c_1 = c_2 = 0$ .

**Dependent (too many vectors).**  $\mathbf{e}_1 = (1, 0)$ ,  $\mathbf{e}_2 = (0, 1)$ , and  $\mathbf{u} = (1, 1)$ :

$$\mathbf{u} = \mathbf{e}_1 + \mathbf{e}_2 \quad \Longleftrightarrow \quad \mathbf{e}_1 + \mathbf{e}_2 - \mathbf{u} = \mathbf{0}.$$

Coefficients  $(1, 1, -1)$  are not all zero, so these three vectors are **linearly dependent**. A basis of  $\mathbb{R}^2$  has exactly 2 vectors, not 3.

### 3.3 Linear dependence — columns example

**Example (columns of  $A$ ).**  $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ , columns  $\mathbf{a}_1 = (1, 2)^T$ ,  $\mathbf{a}_2 = (2, 4)^T$ .

**Same direction.**  $\mathbf{a}_2 = 2\mathbf{a}_1$ , so the two columns lie on one line through the origin.

**Nontrivial combination to 0.**

$$2\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{0} \iff 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Coefficients  $(2, -1)$  are not both zero, so  $\mathbf{a}_1, \mathbf{a}_2$  are **linearly dependent** (only one independent direction in  $\text{Col}(A)$ ).



### 3.3 Solving $A\mathbf{x} = \mathbf{b}$ — example

**Example.**  $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ , so  $\text{Col}(A) = \text{span}\{(1, 2)\}$  (a line in  $\mathbb{R}^2$ ).

**Solvable.**  $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3(1, 2)$ , so  $A\mathbf{x} = \mathbf{b}$  has solutions (e.g.  $\mathbf{x} = (1, 1)^T$ ; not unique).

**Not solvable (existence fails).**  $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \notin \text{Col}(A)$ , so  $A\mathbf{x} = \mathbf{b}$  has **no** solution.

**Dependent columns (uniqueness fails).**  $\mathbf{a}_2 = 2\mathbf{a}_1$ , so the columns are **linearly dependent**. When a solution exists, it is not unique (e.g. both  $\mathbf{x} = (3, 0)^T$  and  $\mathbf{x} = (1, 1)^T$  give  $\mathbf{b} = (3, 6)^T$ ).

### 3.3 Full basis requirement — only when $m = n$

**Goal.** When does **every**  $\mathbf{b} \in \mathbb{R}^m$  satisfy  $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$  in **exactly one way**?

**Claim.** For  $A \in \mathbb{R}^{m \times n}$ , this happens **only if**  $m = n$ .

**Proof sketch.**

- **Uniqueness**  $\Rightarrow$  columns independent. In  $\mathbb{R}^m$  at most  $m$  vectors can be independent, so  $n \leq m$ .
- **Existence for all  $\mathbf{b}$**   $\Rightarrow \text{Col}(A) = \mathbb{R}^m$ . With only  $n$  columns,  $\dim(\text{Col}(A)) \leq n$ , so  $n \geq m$ .
- Therefore  $m = n$ .

If  $n < m$ , columns cannot span  $\mathbb{R}^m$ ; if  $n > m$ ,  $n$  columns in  $\mathbb{R}^m$  cannot be independent.

### 3.3 Square case — basis = existence + uniqueness

**Now**  $A \in \mathbb{R}^{n \times n}$ ,  $\mathbf{b} \in \mathbb{R}^n$ . Requirement for a basis  $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ : every  $\mathbf{b} \in \mathbb{R}^n$  is  $\mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$  in exactly one way.

**Equivalent to**  $A\mathbf{x} = \mathbf{b}$ .

$$A\mathbf{x} = \mathbf{b} \iff \mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n.$$

**Columns form a basis of  $\mathbb{R}^n$**   $\iff$  existence and uniqueness for **every  $\mathbf{b}$**   $\iff A\mathbf{x} = \mathbf{b}$  has a **unique** solution for every  $\mathbf{b} \in \mathbb{R}^n$ .

## 3.4 Two views of a matrix $A$

**View 1 —  $\mathbf{x}$  as weights.** Write  $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$ . Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

The entries of  $\mathbf{x}$  are the **weights** on the columns of  $A$ .

**View 2 —  $A$  as a linear transformation.** Define  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  by  $T(\mathbf{x}) = A\mathbf{x}$ . The matrix  $A$  sends each input vector  $\mathbf{x}$  to an output vector in  $\mathbb{R}^m$ ; for square  $A$ ,  $|\det(A)|$  is the area-scaling factor (Section 3.5).

## 3.4 Linear transformation — definition

**Linear transformation.** A map  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is **linear** if for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and all scalars  $c \in \mathbb{R}$ :

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}), \quad T(c\mathbf{x}) = cT(\mathbf{x}).$$

Equivalently, for all  $\mathbf{x}, \mathbf{y}$  and  $c$ :

$$T(c\mathbf{x} + \mathbf{y}) = cT(\mathbf{x}) + T(\mathbf{y}).$$

**Examples.** Rotation, reflection, projection, and shear in the plane are linear. A map with a constant term, e.g.  $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$  with  $\mathbf{b} \neq \mathbf{0}$ , is **not** linear.

**Matrix form.** Every linear  $T$  is  $T(\mathbf{x}) = A\mathbf{x}$  for some matrix  $A$  (general recipe on the next slides). For  $A \in \mathbb{R}^{2 \times 2}$ ,  $|\det(A)|$  is the area-scaling factor (Section 3.5).

## 3.4 Linear maps — decompose, then combine

**Benefit.** A linear map  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  may be hard to visualize on an **arbitrary** input  $\mathbf{v}$ .

**But you often know  $T$  on a few vectors.** Suppose you already know  $T(\mathbf{x})$  and  $T(\mathbf{y})$ , and  $\mathbf{v}$  can be decomposed as

$$\mathbf{v} = a\mathbf{x} + b\mathbf{y}$$

for some scalars  $a, b \in \mathbb{R}$ . Then by linearity,

$$T(\mathbf{v}) = T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}).$$

**Decompose  $\mathbf{v}$  into known pieces, transform each piece, then combine.**

**Standard basis recipe ( $\mathbb{R}^2$ ).** Take  $\mathbf{x} = \mathbf{e}_1$ ,  $\mathbf{y} = \mathbf{e}_2$ , so  $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ . If you know  $T(\mathbf{e}_1)$  and  $T(\mathbf{e}_2)$ , then  $T(\mathbf{v}) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2)$ . Stack  $T(\mathbf{e}_1)$  and  $T(\mathbf{e}_2)$  as columns to get the matrix of  $T$  (next slide).

## 3.4 Decompose, transform, combine — notation

**Recipe.** For  $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$  and linear  $T$ :

1. **Decompose** the input:

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2.$$

2. **Transform** each piece (use known values of  $T$  on  $\mathbf{e}_1, \mathbf{e}_2$ ):

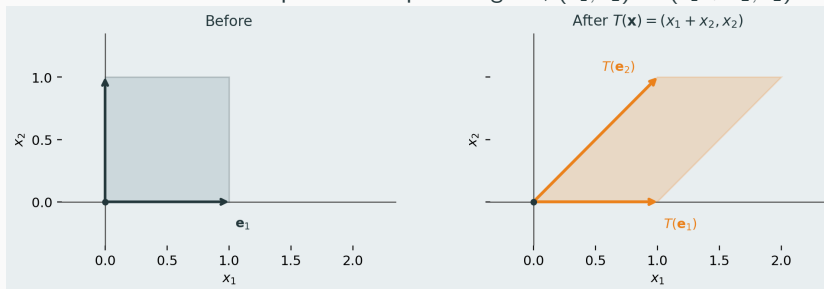
$$v_1 T(\mathbf{e}_1), \quad v_2 T(\mathbf{e}_2).$$

3. **Combine** the outputs:

$$T(\mathbf{v}) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) = A\mathbf{v}, \quad A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix}.$$

## 3.4 Example — shear

**Map.** Horizontal shear: tilt the unit square into a parallelogram,  $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$ .





### 3.4 Example — shear (matrix)

**Find the matrix.** Apply  $T$  to  $\mathbf{e}_1$  and  $\mathbf{e}_2$ , then stack the outputs as columns:

$$T(\mathbf{e}_1) = \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad T(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \Rightarrow \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check:  $A\mathbf{e}_1 = \mathbf{e}_1$  (bottom edge fixed);  $A\mathbf{e}_2$  tilts the vertical side.

### 3.4 Example — shear (decompose, transform, combine)

Take  $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ .

1. **Decompose:**  $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$ .

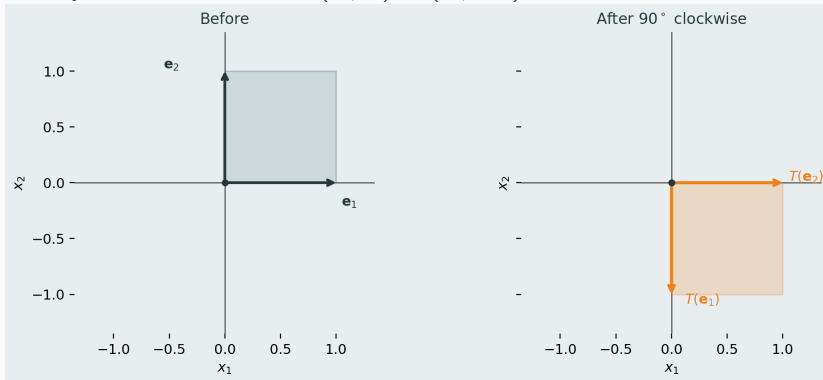
2. **Transform:**  $2T(\mathbf{e}_1) = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}.$

3. **Combine:**

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} = A\mathbf{v}.$$

### 3.4 Example — $90^\circ$ rotation (clockwise)

**Map.** Rotate every vector  $90^\circ$  clockwise:  $(x_1, x_2) \mapsto (x_2, -x_1)$ .



### 3.4 Example — rotation (matrix)

**Find the matrix.** Apply  $T$  to  $\mathbf{e}_1$  and  $\mathbf{e}_2$ , then stack the outputs as columns:

$$T(\mathbf{e}_1) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad T(\mathbf{e}_2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \implies A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Check:  $A\mathbf{e}_1$  points down;  $A\mathbf{e}_2$  points right (see previous graph).

### 3.4 Example — rotation (decompose, transform, combine)

Same recipe with  $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ .

1. **Decompose:**  $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$ .

2. **Transform:**  $2T(\mathbf{e}_1) = 2 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$

3. **Combine:**

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} = A\mathbf{v}.$$

## 3.4 Every linear map is a matrix

**Claim.** If  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is linear, then there is a unique matrix  $A \in \mathbb{R}^{m \times n}$  such that  $T(\mathbf{x}) = A\mathbf{x}$  for all  $\mathbf{x}$ .

**Construction.** Let  $\mathbf{e}_1, \dots, \mathbf{e}_n$  be the standard basis of  $\mathbb{R}^n$ . Form  $A$  by stacking the outputs as columns:

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)].$$

**Why it works.** Any  $\mathbf{x} = \sum_{j=1}^n x_j \mathbf{e}_j$ . By linearity,

$$T(\mathbf{x}) = \sum_{j=1}^n x_j T(\mathbf{e}_j) = \sum_{j=1}^n x_j \mathbf{a}_j = A\mathbf{x}.$$

Column  $j$  of  $A$  is exactly where  $\mathbf{e}_j$  is sent.

### 3.5 Determinant — area scaling ( $2 \times 2$ )

**Setup.**  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , columns  $\mathbf{a}_1 = (a, c)^T$ ,  $\mathbf{a}_2 = (b, d)^T$ . View  $A$  as the linear map  $T(\mathbf{x}) = A\mathbf{x}$  (shear, rotation, etc. on previous slides).

**Where the standard basis goes.**

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \mathbf{a}_1, \quad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \mathbf{a}_2.$$

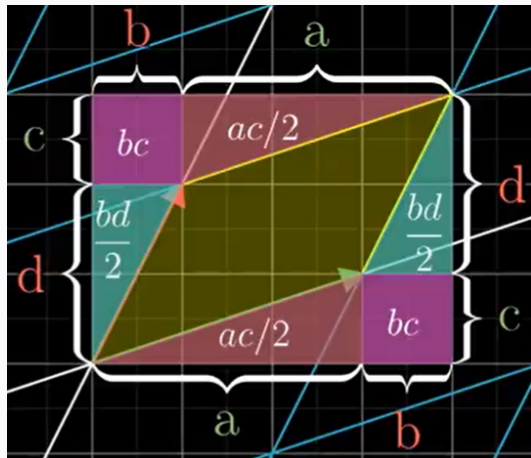
So  $(1, 0)^T$  is sent to  $(a, c)^T$ , and  $(0, 1)^T$  to  $(b, d)^T$ .

**Area conversion factor.** The unit square (spanned by  $(1, 0)^T$  and  $(0, 1)^T$ , area = 1) is mapped by  $A$  to the parallelogram spanned by  $\mathbf{a}_1$  and  $\mathbf{a}_2$ . The **determinant** is exactly the (signed) area of that parallelogram:

$$\det(A) = ad - bc.$$

So  $\det(A)$  is the **area conversion factor**: every region's area is multiplied by  $|\det(A)|$  under  $\mathbf{x} \mapsto A\mathbf{x}$ .

### 3.5 Determinant — why $ad - bc$ is the area



Box area  $(a + b)(c + d)$  minus purple  $(bc)$ , red  $(ac/2)$ , teal  $(bd/2)$  pieces leaves yellow parallelogram area

$$ad - bc = \det(A).$$



## 3.5 Determinant = 0 — example

**Example.**  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , so  $a = b = c = d = 1$  and

$$\det(A) = ad - bc = 1 \cdot 1 - 1 \cdot 1 = 0.$$

**Standard basis collapses to a line.**

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both  $(1, 0)^T$  and  $(0, 1)^T$  are sent to the **same** point on one line through the origin. The unit square (area 1) is squashed to a **1-dimensional** set — area 0.

**So (square case):**

$$\det(A) = 0 \iff \text{columns linearly dependent.}$$

Here  $\mathbf{a}_2 = \mathbf{a}_1 = (1, 1)^T$ , so the columns are dependent. **Next:** when  $\det(A) = 0$ , some input direction is **squeezed** to **0**.

## 3.5 Null space — directions squeezed to 0

**Setup** ( $A \in \mathbb{R}^{2 \times 2}$ ). View  $A$  as a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $T(\mathbf{x}) = A\mathbf{x}$ .

**Null space / kernel** (subspace of  $\mathbb{R}^2$ ):

$$\text{Null}(A) = \ker(T) = \{\mathbf{x} \in \mathbb{R}^2 : A\mathbf{x} = \mathbf{0}\}.$$

Vectors in  $\text{Null}(A)$  are mapped to  $\mathbf{0}$  in the output.

**Intuition.** If  $\mathbf{x} \neq \mathbf{0}$  lies in  $\text{Null}(A)$ , the whole line  $\text{span}\{\mathbf{x}\} \subset \mathbb{R}^2$  is sent to the single point  $\mathbf{0} \in \mathbb{R}^2$  — that **input direction is squeezed** (lost in the output).

**Example** ( $\det A = 0$ ).  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ :  $A(1, -1)^T = \mathbf{0}$ , so  $\text{Null}(A) = \text{span}\{(1, -1)^T\}$  — one direction squeezed.

**Next:** how many directions squeeze vs. persist?

### 3.5 Rank–nullity — squeezed vs. persist ( $2 \times 2$ )

**Two fates** ( $A \in \mathbb{R}^{2 \times 2}$ ). Under  $T(\mathbf{x}) = A\mathbf{x}$ , each input direction either:

1. **Squeezed to 0**  $\ker(T) = \text{Null}(A)$  (previous slide); dimension = nullity( $A$ ).
2. **Persist in the output**  $\text{Im}(T) = \text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^2\}$ ; dimension = rank( $A$ ) — independent directions that **survive**.

**Rank–nullity.**  $\text{rank}(A) + \text{nullity}(A) = 2$ .

**Example** ( $\det A = 0$ ). Collapse matrix from previous slides: nullity = 1, rank = 1 (outputs fill  $\text{span}\{(1, 1)^T\}$ ).

**Example** ( $\det A \neq 0$ ). Shear or rotation (Section 3.4): nullity = 0, rank = 2.

### 3.5 After a squeeze — what survives? ( $2 \times 2$ )

**Recall.**  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  maps the whole plane onto one output line (through  $(1, 1)^T$ ). Direction  $(1, -1)^T$  is squeezed to  $\mathbf{0}$ .

**Question.** Given an input  $\mathbf{v}$ , what is  $A\mathbf{v}$ ?

**Key idea (in words).** Think of  $\mathbf{v}$  as two moves: one along the **squeezed** direction (the map **forgets** it) and one along the surviving direction  $(1, 1)^T$  (the map **remembers** only this). So  $A\mathbf{v}$  depends on how much of  $\mathbf{v}$  points along the surviving direction — not on the part that collapses.

**Example.**  $(3, 1)^T = 2(1, 1)^T + 1(1, -1)^T$ : the  $(1, -1)^T$  part is lost, the  $2(1, 1)^T$  part survives, so  $A\mathbf{v} = 2(2, 2)^T = (4, 4)^T$ .

**Next:** the same squeeze/persist picture in general  $m \times n$ .

### 3.5 Rank–nullity — general case **for interest only**

**Setup.** Let  $A \in \mathbb{R}^{m \times n}$ ,  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ,  $T(\mathbf{x}) = A\mathbf{x}$ .

**Kernel and image.**  $\ker(T) = \text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$ ;  
 $\text{Im}(T) = \text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$ .

**Rank and nullity.**  $\text{rank}(A) = \dim(\text{Im } T)$ ,  $\text{nullity}(A) = \dim(\ker T)$ .

**Rank–nullity theorem.**

$$\text{rank}(A) + \text{nullity}(A) = n \quad (n \text{ columns} = \dim(\mathbb{R}^n)).$$

**Same intuition.** Of the  $n$  domain dimensions,  $\text{nullity}(A)$  are **squeezed** to  $\mathbf{0} \in \mathbb{R}^m$ ;  $\text{rank}(A)$  independent directions **persist** in  $\text{Im}(T) \subseteq \mathbb{R}^m$ . When  $n = m$ ,  $\det(A) \neq 0 \Leftrightarrow \text{nullity} = 0 \Leftrightarrow \text{rank} = n$  (same as invertible / basis on the next slide).

## 3.5 Invertible matrices — column picture

Recall ( $A \in \mathbb{R}^{n \times n}$ ).  $Ax = b \iff b = x_1 a_1 + \cdots + x_n a_n$  (§3.3).

**Basis = two requirements (§3.3).** Columns  $a_1, \dots, a_n$  form a **basis** of  $\mathbb{R}^n$  iff:

- (1) **Uniqueness:** at most one choice of weights  $x_1, \dots, x_n$  (columns **linearly independent**).
- (2) **Solvability:** every  $b \in \mathbb{R}^n$  is such a combination ( $\text{Col}(A) = \mathbb{R}^n$ ).

For square  $A$ , (1) $\Leftrightarrow$ (2); together they give a **unique** solution for **every**  $b$ , and  $A^{-1}$  exists.

If (1) or (2) fails (not a basis):

- (2) fails:  $\text{Col}(A) \subsetneq \mathbb{R}^n$  — some  $b$  have **no** solution.
- (1) fails (columns **linearly dependent**):  $\exists d \neq 0$  with  $Ad = 0$  — if a solution exists,  $x + d$  is another (not unique).

**Link to geometry (this section).**  $\det(A) = 0 \Leftrightarrow$  a squeeze  $\Leftrightarrow$  not invertible;  $\det(A) \neq 0 \Leftrightarrow$  nullity = 0  $\Leftrightarrow$  basis  $\Leftrightarrow$  invertible.

## 3.5 Invertible matrices — equivalent conditions

Let  $A \in \mathbb{R}^{n \times n}$ . All eight statements are **equivalent**.

### I. Column picture (§3.3)

1. Columns **linearly independent** — **uniqueness**.
2.  $\text{Col}(A) = \mathbb{R}^n$  — **solvability for all  $\mathbf{b}$** .
3. Columns form a **basis** — (1) and (2) [square:  $(1) \Leftrightarrow (2)$ ].

### II. Solving $A\mathbf{x} = \mathbf{b}$

4.  $A\mathbf{x} = \mathbf{0}$  only  $\mathbf{x} = \mathbf{0}$ .
5. For every  $\mathbf{b}$ , **unique** solution  $\mathbf{x} = A^{-1}\mathbf{b}$ .

### III. Matrix / geometry

6.  $A$  **invertible** ( $AA^{-1} = I_n$ ).
7.  $\det(A) \neq 0$  (no squeeze).
8. Rows form a **basis** of  $\mathbb{R}^n$ .

## 3.5 Transpose and inverse — rules we use

**Transpose (always defined).**

$$(A^T)^T = A, \quad (AB)^T = B^T A^T, \quad (Ax)^T = x^T A^T.$$

**Why (8)?** (2)  $\Rightarrow Ax = \mathbf{b}$  for all  $\mathbf{b}$ . Transpose:  $x^T A^T = \mathbf{b}^T$  for all  $\mathbf{b}^T$  (row view, §3.3). Rows of  $A =$  columns of  $A^T \Rightarrow$  rows span  $\mathbb{R}^n$ ; square  $\Rightarrow$  basis [(8)].

**Inverse (definition).** For square  $A \in \mathbb{R}^{n \times n}$ ,  $A^{-1}$  satisfies  $AA^{-1} = A^{-1}A = I_n$ . It exists iff any condition on the previous slide holds (e.g.  $\det(A) \neq 0$ ). Then  $Ax = \mathbf{b}$  has unique solution  $\mathbf{x} = A^{-1}\mathbf{b}$ .

**Inverse rules (both factors invertible).**

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^{-1})^{-1} = A, \quad (A^{-1})^T = (A^T)^{-1}.$$

Proofs: [appendix](#) (for interest only).



## 3.6 Dot product — algebraic and geometric

For  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ :

**Algebraic definition.**

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

**Geometric definition.** Let  $\theta$  be the angle between  $\mathbf{u}$  and  $\mathbf{v}$ . Then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

**Equivalent.** These two definitions always give the **same number**.

**Video (intuition and proof).** <https://www.youtube.com/watch?v=LyGKycYT2v0>

## 3.6 Linear regression — $y$ and $X$

**Data** ( $n$  workers). Worker  $i$  has income  $y_i$ , age  $\text{age}_i$ , education  $\text{educ}_i$ .

**Response vector.**

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n.$$

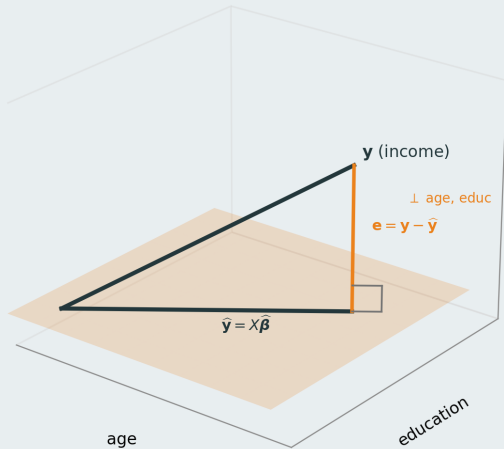
**Regressor matrix** (two columns: age, educ).

$$X = \begin{pmatrix} \text{age}_1 & \text{educ}_1 \\ \vdots & \vdots \\ \text{age}_n & \text{educ}_n \end{pmatrix} \in \mathbb{R}^{n \times 2}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_{\text{age}} \\ \beta_{\text{educ}} \end{pmatrix}, \quad \mathbf{y} \approx X\boldsymbol{\beta}.$$

Column  $j$  of  $X$  is the vector of worker  $i$  values for regressor  $j$ .

Schematic next ( $n = 3$ ): age and education are the two axes; yellow plane =  $\text{Col}(X)$ ;  $\mathbf{e}$  is orthogonal to every column of  $X$ .

Yellow plane  
=  
 $\text{Col}(X)$



## 3.6 OLS fit — $\mathbf{e}$ orthogonal to every column of $X$

**From the picture** ( $n = 3$ ). Yellow plane =  $\text{Col}(X)$ ; fitted  $\hat{\mathbf{y}} = X\hat{\boldsymbol{\beta}}$  lies in it. Residual  $\mathbf{e} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$  is perpendicular to the plane.

**Orthogonal to every column of  $X$ .** Let  $\mathbf{x}_1 = \text{age column}$ ,  $\mathbf{x}_2 = \text{educ column}$ . Then

$$\mathbf{x}_1 \cdot \mathbf{e} = 0, \quad \mathbf{x}_2 \cdot \mathbf{e} = 0 \quad \Longleftrightarrow \quad X^T \mathbf{e} = \mathbf{0} \quad \Longleftrightarrow \quad X^T (\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

These **normal equations** characterise  $\hat{\boldsymbol{\beta}}$ .

**Solve for  $\hat{\boldsymbol{\beta}}$ .**  $X^T X \hat{\boldsymbol{\beta}} = X^T \mathbf{y}$ . If the columns of  $X$  are independent (full column rank), then

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}.$$

(Since  $\hat{\mathbf{y}} \in \text{Col}(X)$ , also  $\hat{\mathbf{y}} \cdot \mathbf{e} = 0$ .)

## 3.6 OLS fit — same as least squares

**Pythagoras in  $\mathbb{R}^n$ .** Fix  $\hat{\beta}$  with  $\mathbf{e} = \mathbf{y} - X\hat{\beta}$ . For any  $\beta$ ,

$$\mathbf{y} - X\beta = \underbrace{\mathbf{y} - X\hat{\beta}}_{\mathbf{e}} + \underbrace{X(\hat{\beta} - \beta)}_{\in \text{Col}(X)}.$$

The two pieces are orthogonal, so

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2 \geq \|\mathbf{y} - X\hat{\beta}\|^2.$$

**Conclusion.** The smallest squared error occurs when  $X(\hat{\beta} - \beta) = \mathbf{0}$ , i.e.  
 $\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - X\beta\|^2 = \arg \min_{\beta} \sum_{i=1}^n (y_i - \beta_{\text{age}} \text{age}_i - \beta_{\text{educ}} \text{educ}_i)^2.$

## 3.6 The projection matrix

**Projection matrix  $P_X$  (hat matrix).** If  $X \in \mathbb{R}^{n \times k}$  has full column rank, define

$$P_X = X(X^T X)^{-1} X^T.$$

Then  $\hat{\mathbf{y}} = P_X \mathbf{y}$ .

**Annihilator matrix  $M_X$  (residual maker matrix).** Define  $M_X = I - P_X$ . Then  $\mathbf{e} = M_X \mathbf{y}$ ; and  $M_X X = \mathbf{0}$  ( $M_X$  annihilates  $\text{Col}(X)$ ).

**Useful facts.**  $P_X^T = P_X$ ,  $P_X^2 = P_X$ ,  $M_X X = \mathbf{0}$ , and  $M_X^T = M_X$ ,  $M_X^2 = M_X$ .

Why  $P_X^T = P_X$  and  $P_X^2 = P_X$ : [appendix \(for interest only\)](#).

### 3.6 Frisch–Waugh–Lovell (FWL) theorem for interest only

**Partition regressors.** Write  $X = [X_1 \mid X_2]$  (e.g. age =  $X_1$ , educ =  $X_2$ ). Let  $P_{X_2}$  be the **projection matrix** onto  $\text{Col}(X_2)$  and  $M_{X_2} = I - P_{X_2}$  the **annihilator matrix** (residual maker matrix; write  $P_2$ ,  $M_2$  for short).

**Claim.** The OLS coefficient on  $X_1$  in  $y \sim X_1 + X_2$  equals the OLS coefficient in

$$M_2 y \sim M_2 X_1$$

( $M_2$  removes the part of each vector explained by educ).

**Reading.** “Controlling for educ” = apply the annihilator  $M_2$  to both income and age, then regress.

**Proof idea.** Write  $y = X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + e$  with  $X^T e = 0$ . Since  $M_2 X_2 = 0$  and  $M_2 e = e$ , get  $M_2 y = M_2 X_1 \hat{\beta}_1 + e$ ; also  $(M_2 X_1)^T e = X_1^T M_2 e = X_1^T e = 0$ , so  $\hat{\beta}_1$  solves the partialled normal equations.

### 3.7 Which basis makes decompose–transform–combine easy?

**Recall the recipe.** If  $\mathbf{v} = a\mathbf{x} + b\mathbf{y}$ , then  $T(\mathbf{v}) = aT(\mathbf{x}) + bT(\mathbf{y})$  — always true by linearity.

**But step 2 can be hard.** You must know  $T(\mathbf{x})$  and  $T(\mathbf{y})$ . With the standard basis  $(\mathbf{e}_1, \mathbf{e}_2)$ , rotation sent  $\mathbf{e}_1$  down and  $\mathbf{e}_2$  right — both outputs are **new directions** to track.

**A proper basis (for this map).** Pick directions where  $T$  **only scales**, not rotates or shears:

$$T(\mathbf{v}) = \lambda\mathbf{v} \quad \text{for some scalar } \lambda.$$

Then  $T(a\mathbf{v}) = a\lambda\mathbf{v}$  — one number  $\lambda$ , same direction. These special directions are **eigenvectors**;  $\lambda$  is the **eigenvalue**.



## 3.7 Eigenvectors and eigenvalues — definition

**Setup.** Square matrix  $A \in \mathbb{R}^{n \times n}$  (linear map  $T(\mathbf{x}) = A\mathbf{x}$ ).

**Eigenvector / eigenvalue.** A nonzero vector  $\mathbf{v}$  is an **eigenvector** of  $A$  if

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar  $\lambda$  (the **eigenvalue**). Geometrically:  $A$  stretches or flips  $\mathbf{v}$ , but keeps its line.

**Proper basis.** When  $A$  has enough independent eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$ , decompose in **that** basis instead of the standard one.

If  $A = A^T$ , eigenvalues are real (**appendix**); if also PSD,  $\lambda_i \geq 0$  (**appendix**) (**for interest only**).

## 3.7 Finding eigenvalues and eigenvectors

**Traditional recipe** (works in any dimension):

1. **Eigenvalues.** Solve the **characteristic equation**

$$\det(A - \lambda I) = 0.$$

2. **Eigenvectors.** For each eigenvalue  $\lambda$ , solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Nontrivial solutions are eigenvectors; the solution set is the **eigenspace**  $E_\lambda$ .

**Small**  $n$  (e.g.  $2 \times 2$ ): solve the equations by hand (Together exercise below).

**General / high**  $n$ : row-reduce  $(A - \lambda I)$  to **RREF** and read off pivot vs. free columns — **appendix (for interest only)**.

3. **Repeated**  $\lambda$ . If  $\lambda$  has algebraic multiplicity  $> 1$ ,  $\dim E_\lambda$  equals the number of free variables in step 2.

Together:  $2 \times 2$  by hand. Take home:  $\mathbb{R}^4$  exercise below.

### 3.7 Exercise: Eigenvectors — $2 \times 2$ example Together

**Exercise.** For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , find all eigenpairs and compute  $A\mathbf{v}$  for  $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ .

**Solution (eigenvalues).** Characteristic equation  $\det(A - \lambda I) = 0$ :

$$(2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0 \quad \Rightarrow \quad \lambda_1 = 3, \lambda_2 = 1.$$

**Solution ( $\lambda_1 = 3$ ).** Solve  $(A - 3I)\mathbf{v} = \mathbf{0}$ :

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad -v_1 + v_2 = 0 \quad \Rightarrow \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

**Solution ( $\lambda_2 = 1$ ).** Solve  $(A - I)\mathbf{v} = \mathbf{0}$ :

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad v_1 + v_2 = 0 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Check:  $A\mathbf{v}_1 = 3\mathbf{v}_1$ ,  $A\mathbf{v}_2 = 1\mathbf{v}_2$ .

### 3.7 Exercise: Eigenvectors — $2 \times 2$ example (cont.)

Together

**Compute**  $A\mathbf{v}$  for  $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$  (decompose–transform–combine).

**Step 1 (decompose).** Write  $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ :

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \Rightarrow c_1 = 2, c_2 = 1.$$

**Step 2 (transform).**  $A$  scales each eigen-direction by its eigenvalue:

$$A\mathbf{v} = 2 \cdot 3\mathbf{v}_1 + 1 \cdot 1\mathbf{v}_2 = \begin{pmatrix} 7 \\ 5 \end{pmatrix}.$$

### 3.7 Exercise: Eigenvectors — $2 \times 2$ example (cont.)

Together

**Compute**  $A^{10}\mathbf{v}$  for  $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2\mathbf{v}_1 + \mathbf{v}_2$ .

On an eigenvector:  $A^{10}\mathbf{v}_1 = A^9(A\mathbf{v}_1) = A^9(\lambda_1\mathbf{v}_1) = \lambda_1 A^9\mathbf{v}_1 = \cdots = \lambda_1^{10}\mathbf{v}_1$  (and similarly  $A^{10}\mathbf{v}_2 = \lambda_2^{10}\mathbf{v}_2$ ).

Same recipe: multiply each coefficient by  $\lambda_i^{10}$ :

$$A^{10}\mathbf{v} = 2 \cdot 3^{10}\mathbf{v}_1 + 1 \cdot 1^{10}\mathbf{v}_2.$$

Since  $3^{10} = 59049$ ,

$$A^{10}\mathbf{v} = 118098 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 118099 \\ 118097 \end{pmatrix}.$$

The  $\lambda_1 = 3$  term dominates; repeated multiplication amplifies the  $(1, 1)$  direction.

### 3.7 Exercise: Eigenvectors in $\mathbb{R}^4$ (RREF) Take home

**Exercise.** For

$$A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix},$$

find all eigenvalues. For  $\lambda = 2$ , row-reduce  $(A - 2I)$  to RREF and find a basis of  $E_2$  (RREF: [appendix](#)).

**Solution (eigenvalues).** Two decoupled  $2 \times 2$  blocks; each gives  $(3 - \lambda)^2 - 1 = 0$ .

$$\det(A - \lambda I) = ((3 - \lambda)^2 - 1)^2 = (\lambda - 2)^2(\lambda - 4)^2.$$

So  $\lambda = 2$  and  $\lambda = 4$ , each with algebraic multiplicity 2.

### 3.7 Exercise: Eigenvectors in $\mathbb{R}^4$ (RREF, cont.)

Take home

**Solution** ( $\lambda = 2$ ). Row-reduce

$$A - 2I = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

**Pivot columns** 1, 3  $\Rightarrow v_1, v_3$  pivot; **free columns** 2, 4  $\Rightarrow v_2, v_4$  free. Equations:  $v_1 + v_2 = 0$ ,  $v_3 + v_4 = 0$ .

**General solution:**  $(v_1, v_2, v_3, v_4) = v_2(-1, 1, 0, 0) + v_4(0, 0, -1, 1)$ .

**Basis of  $E_2$**  (Appendix A.6 recipe):  $(-1, 1, 0, 0)^\top, (0, 0, -1, 1)^\top$ .

## 3.7 Eigenbasis — especially useful in high dimension

If  $\mathbf{v}_1, \dots, \mathbf{v}_n$  is an eigenbasis and  $\mathbf{v} = \sum_i c_i \mathbf{v}_i$ , then

$$A\mathbf{v} = \sum_i c_i \lambda_i \mathbf{v}_i, \quad A^k \mathbf{v} = \sum_i c_i \lambda_i^k \mathbf{v}_i.$$

In an eigenbasis,  $A$  is just **separate scalings** — easy to describe and compute when  $n$  is large.

Not every matrix has a full real eigenbasis (e.g. plane rotation); **appendix (for interest only)**. SVD/QR play a similar role when it does not.



## 3.7 Applications — PCA and Markov chains **for interest only**

**for interest only** You may need to understand some of these when you pursue macro, empirical economics, or IO.

**Extremely useful beyond the  $2 \times 2$  example.** Many applied problems reduce to: find directions where a matrix acts simply.

- **PCA (principal components).** Eigenvectors of the **covariance matrix** are the main directions of variation in data; largest eigenvalues pick the most important components for dimension reduction (**appendix**).
- **Markov transition matrices.** For a stochastic matrix  $P$  (rows sum to 1), the **long-run distribution** is an eigenvector with  $\lambda = 1$ ; other eigenvalues control how fast  $P^k \mathbf{v}$  converges to that steady state.

**Pattern.** Decompose a state or data vector in special directions; each direction evolves by a single number ( $\lambda_i$ ).

## 3.7 Applications — ODEs for interest only

Ordinary differential equations (ODEs) — eigenvalues / eigenvectors show up in:

- Linear systems  $\dot{x} = Ax$
- Stability of equilibria and phase-plane dynamics
- Continuous-time macro and growth models
- Linearized dynamics around steady states
- Compartment models (e.g. epidemiology, population dynamics)

## 3.7 Decompositions — QR, eigen, SVD for interest only

Different decompositions, same spirit: rewrite a matrix so that the hard part is **diagonal**, **triangular**, or **orthogonal**.

- **QR decomposition** ( $A = QR$ ).  $Q$  has orthonormal columns spanning  $\text{Col}(A)$ ;  $R$  is upper triangular. Used to pick **linearly independent columns**, solve least squares stably, and build regression numerically.
- **Spectral / eigen decomposition** ( $A = V\Lambda V^{-1}$ ). When  $A$  has a full eigenbasis,  $V$  lists eigenvectors and  $\Lambda$  lists eigenvalues — matrix powers and dynamics become coefficient-wise.
- **Singular Value Decomposition (SVD)** ( $A = U\Sigma V^T$ ). Always exists; built from eigenproblems on  $A^T A$  (and  $AA^T$ ). Underlies PCA, low-rank approximation, and regularized regression.

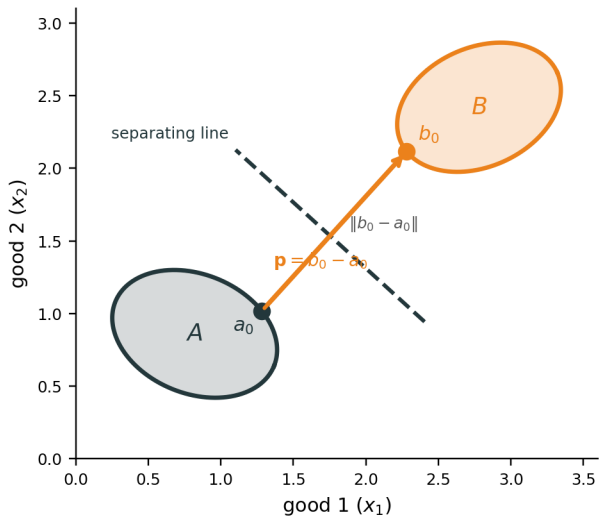
Not required for Math Camp — names and connections only.

Self-study video. <https://www.youtube.com/watch?v=PFDu9oVAE-g>

# Appendix

**Not required for Math Camp.** Proofs and extra examples linked from the main slides.

Proof idea: closest pair, then a line  $\perp \mathbf{p}$



## A.1 Proof. Separation when $A$ is compact

for interest only

**Claim.** Disjoint nonempty convex  $A, B \subset \mathbb{R}^n$  with  $A$  compact and  $B$  closed. Then  $\exists \mathbf{p} \neq \mathbf{0}, \beta \in \mathbb{R}$  with  $\mathbf{p} \cdot \mathbf{a} \leq \beta \leq \mathbf{p} \cdot \mathbf{b}$  for all  $\mathbf{a} \in A, \mathbf{b} \in B$ .

**Step 1 — closest pair.** Define  $d = \inf\{\|\mathbf{a} - \mathbf{b}\| : \mathbf{a} \in A, \mathbf{b} \in B\} > 0$ . Pick sequences  $(\mathbf{a}_n) \subset A, (\mathbf{b}_n) \subset B$  with  $\|\mathbf{a}_n - \mathbf{b}_n\| \rightarrow d$ . Since  $A$  is compact, a subsequence  $\mathbf{a}_{n_k} \rightarrow \mathbf{a}_0 \in A$ . The  $(\mathbf{b}_{n_k})$  are bounded, so (using  $B$  closed) a further subsequence  $\mathbf{b}_{n_{k_j}} \rightarrow \mathbf{b}_0 \in B$ . Hence the infimum is **attained** at some pair  $(\mathbf{a}_0, \mathbf{b}_0) \in A \times B$ .

**Not an interior point.**  $\mathbf{a}_0$  and  $\mathbf{b}_0$  need only lie in  $A$  and  $B$ ; they are usually on the **boundaries**, not in  $\text{int}(A)$  or  $\text{int}(B)$ .

**Step 2 — normal vector.** Set  $\mathbf{p} = \mathbf{b}_0 - \mathbf{a}_0 \neq \mathbf{0}$  (points from  $\mathbf{a}_0$  toward  $\mathbf{b}_0$ ).

## A.1 Step 1 — why the subsequence argument works

for interest only

**Why  $a_{n_k} \rightarrow a_0 \in A$ .** In  $\mathbb{R}^n$ , **compact** = closed + bounded (Heine–Borel). Every sequence in a compact set has a **subsequence** converging to a limit **in that set**. So  $(a_n) \subset A$  compact  $\Rightarrow \exists a_{n_k} \rightarrow a_0 \in A$ .

**Why  $(b_{n_k})$  is bounded.** Since  $\|a_{n_k} - b_{n_k}\| \rightarrow d$  and  $(a_{n_k})$  converges, for large  $k$ ,

$$\|b_{n_k}\| \leq \|b_{n_k} - a_{n_k}\| + \|a_{n_k}\| \leq d + 1 + \|a_0\| + 1.$$

(Triangle inequality:  $\|b\| \leq \|b - a\| + \|a\|$ .)

**Why  $b_{n_{k_j}} \rightarrow b_0 \in B$ .** A bounded sequence in  $\mathbb{R}^n$  has a convergent subsequence (Bolzano–Weierstrass). Since  $B$  is **closed**, the limit  $b_0$  lies in  $B$ .

## A.1 Step 3 — why the derivative works

for interest only

**Setup.** Fix  $a \in A$ . For  $t \in [0, 1]$ , convexity gives  $a_t = a_0 + t(a - a_0) \in A$  and  $g(t) = \|a_t - b_0\|^2$ . Since  $(a_0, b_0)$  minimizes distance,  $t = 0$  minimizes  $g$  on  $[0, 1]$ .

**Why use the derivative?** A minimum at the left endpoint  $t = 0$  means the slope cannot point downward:  $g'(0) \geq 0$ .

**Differentiate.**  $g(t) = \|a_0 + t(a - a_0) - b_0\|^2$ , so

$$g'(t) = 2\langle a_0 + t(a - a_0) - b_0, a - a_0 \rangle, \quad g'(0) = 2\langle a_0 - b_0, a - a_0 \rangle \geq 0.$$

Hence  $\langle b_0 - a_0, a - a_0 \rangle \leq 0$ . With  $\mathbf{p} = b_0 - a_0$ :  $\mathbf{p} \cdot (a - a_0) \leq 0$ , so  $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$ .



## A.1 Proof. Separation when $A$ is compact (continued)

**Step 4 —  $B$  stays on the other side.** Fix  $b \in B$ . For  $t \in [0, 1]$ , set  $b_t = b_0 + t(b - b_0) \in B$  and  $h(t) = \|a_0 - b_t\|^2$ . Since  $t = 0$  minimizes  $h$  on  $[0, 1]$ , the same derivative argument gives  $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$  for all  $b \in B$ .

**Step 5 — finish.** Since  $A \cap B = \emptyset$ , the closest points satisfy  $a_0 \neq b_0$ . With  $\mathbf{p} = b_0 - a_0$ ,

$$\mathbf{p} \cdot b_0 - \mathbf{p} \cdot a_0 = \mathbf{p} \cdot (b_0 - a_0) = \|b_0 - a_0\|^2 = d^2 > 0,$$

so  $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$ . Any  $\beta$  with  $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$  works (e.g.  $\beta = \mathbf{p} \cdot a_0$  gives **weak** separation:  $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$  for all  $a \in A, b \in B$ ).

## A.1 Strict separation

for interest only

**Weak vs. strict.** From Steps 3–5:  $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$  and  $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$ . So any  $\beta$  with  $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$  separates weakly (e.g.  $\beta = \mathbf{p} \cdot a_0$ ).

**When is separation strict?** If  $d = \|b_0 - a_0\| > 0$  (sets bounded apart; e.g. both compact and disjoint), then  $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$  with a **positive gap**. Pick  $\beta$  **strictly between** them, e.g.  $\beta = \frac{1}{2}(\mathbf{p} \cdot a_0 + \mathbf{p} \cdot b_0)$ . Then

$$\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0 < \beta < \mathbf{p} \cdot b_0 \leq \mathbf{p} \cdot b \quad \Rightarrow \quad \mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b.$$

If  $d = 0$  (sets touch in the limit, e.g.  $A = [0, 1]$ ,  $B = (1, 2]$ ), only weak separation is possible.

## A.1 Proof. Remarks

for interest only

**If only  $B$  is compact.** Swap the roles of  $A$  and  $B$ , apply the proof, then relabel the normal vector.

**Single point  $B = \{b_0\}$ .** Let  $\hat{a}$  be a point in  $A$  closest to  $b_0$  (exists since  $A$  is closed and convex). The same calculation with  $(a_0, b_0) = (\hat{a}, b_0)$  separates  $A$  from  $b_0$  even when  $A$  is not compact.

## A.2 Proof. Efficient production $\Rightarrow$ profit maximization

for interest only

**Goal.**  $y \in Y$  technically efficient  $\Rightarrow \exists \mathbf{p} \neq \mathbf{0}$  with  $\mathbf{p} \cdot y \geq \mathbf{p} \cdot y'$  for all  $y' \in Y$ .

**Assumptions (camp level).**  $Y \subset \mathbb{R}^n$  is **closed and convex**;  $Y$  is **bounded** (so compact); **free disposal** (if  $y \in Y$  and  $\tilde{y} \leq y$ , then  $\tilde{y} \in Y$ ).

**Step 1 — better set.** Define

$$P_y = \{y' \in \mathbb{R}^n : y' \geq y, y' \neq y\}$$

(weakly more output, strictly more in at least one good).  $P_y$  is **convex**.

**Step 2 — disjointness.** If  $y' \in P_y \cap Y$ , then  $y'$  weakly dominates  $y$  in  $Y$ , contradicting efficiency. So  $P_y \cap Y = \emptyset$ .

## A.2 Proof. Efficient production (continued)

for interest only

**Step 3 — separate.**  $Y$  is compact and  $P_y$  is closed, so apply the separating hyperplane theorem (swap labels if needed so a compact set is on one side). There exist  $\mathbf{p} \neq \mathbf{0}$  and  $\beta$  with

$$\mathbf{p} \cdot \mathbf{y}'' \leq \beta \leq \mathbf{p} \cdot \mathbf{y}' \quad \forall \mathbf{y}'' \in Y, \forall \mathbf{y}' \in P_y.$$

Hence  $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot \mathbf{y}'$  for all  $\mathbf{y}'' \in Y, \mathbf{y}' \in P_y$ .

**Step 4 — profit maximization at  $y$ .** For each good  $i$ , pick  $\mathbf{y}^{(i)} = y + \varepsilon \mathbf{e}_i \in P_y$  for small  $\varepsilon > 0$  (feasible with free disposal). Then  $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot \mathbf{y}^{(i)} = \mathbf{p} \cdot y + \varepsilon p_i$ . Let  $\varepsilon \downarrow 0$  to get  $\mathbf{p} \cdot \mathbf{y}'' \leq \mathbf{p} \cdot y$  for all  $\mathbf{y}'' \in Y$ .

**Remark.** Separation gives only  $\mathbf{p} \neq \mathbf{0}$ . For  $\mathbf{p} \geq 0$ , need free disposal on all goods: if  $p_i < 0$ , then  $y - \varepsilon \mathbf{e}_i \in Y$  (since  $\leq y$ ) raises profit, contradicting maximization at  $y$ .

## A.3 Transpose rules — proof

for interest only

$(A^T)^T = A$ . Transpose swaps rows and columns twice, so you return to  $A$ .

$(AB)^T = B^T A^T$ . Entry  $(i, j)$  of  $(AB)^T$  equals entry  $(j, i)$  of  $AB$ :

$$((AB)^T)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}.$$

Same matrix  $\Rightarrow$  equal.

$(A\mathbf{x})^T = \mathbf{x}^T A^T$ . Special case with  $B = \mathbf{x}$  a column  $(n \times 1)$ :  $(A\mathbf{x})^T = \mathbf{x}^T A^T$ .

## A.3 Inverse rules — proof

for interest only

$(AB)^{-1} = B^{-1}A^{-1}$  (both invertible). Multiply in reverse order:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = I.$$

Similarly  $(B^{-1}A^{-1})(AB) = I$ . So  $B^{-1}A^{-1}$  is the inverse of  $AB$ .

$(A^{-1})^{-1} = A$ . By definition of inverse,  $A^{-1}A = I$ , so  $A$  is the inverse of  $A^{-1}$ .

$(A^{-1})^T = (A^T)^{-1}$ . Transpose  $A^{-1}A = I$ :

$$(A^{-1}A)^T = A^T(A^{-1})^T = I.$$

So  $(A^{-1})^T$  is the inverse of  $A^T$ .

## A.4 Projection matrix — why $P_X^2 = P_X$ ?

for interest only

**Setup.**  $P_X = X(X^T X)^{-1} X^T$  is the **projection matrix** (hat matrix);  $M_X = I - P_X$  is the **annihilator matrix** (residual maker matrix).

**Idempotent:**  $P_X^2 = P_X$  (**project once is enough**).  $P_X \mathbf{y} = \hat{\mathbf{y}}$  already lies in  $\text{Col}(X)$ . Vectors in that yellow plane are unchanged by projecting onto it again:

$$P_X \hat{\mathbf{y}} = \hat{\mathbf{y}} \quad \implies \quad P_X(P_X \mathbf{y}) = P_X \mathbf{y}.$$

First drop the part orthogonal to  $\text{Col}(X)$ ; a second drop removes nothing.



## A.4 Dot product and transpose — why $(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T\mathbf{v})$ ?

for interest only

**Goal.** This is the **transpose rule** used in the symmetry proof of  $P_X$ .

**Rewrite both sides as row  $\times$  column.** Dot product = transpose the first vector, then multiply:

$$(A\mathbf{u}) \cdot \mathbf{v} = (A\mathbf{u})^T \mathbf{v} = \mathbf{u}^T A^T \mathbf{v},$$

using  $(A\mathbf{u})^T = \mathbf{u}^T A^T$ .

**Other side is the same expression.**

$$\mathbf{u} \cdot (A^T \mathbf{v}) = \mathbf{u}^T (A^T \mathbf{v}).$$

So  $(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T \mathbf{v})$  for every  $A, \mathbf{u}, \mathbf{v}$ .

Meaning:  $A^T$  moves  $A$  from the left of  $\mathbf{u}$  to the right of  $\mathbf{u}$  inside an inner product.

## A.4 Projection matrix — why $P_X^T = P_X$ ?

for interest only

**Step 1 — geometric balance.**  $P_X \mathbf{u} \in \text{Col}(X)$ , so it is orthogonal to the part of  $\mathbf{v}$  outside that space:

$$(P_X \mathbf{u}) \cdot \mathbf{v} = (P_X \mathbf{u}) \cdot (P_X \mathbf{v}).$$

Swapping  $\mathbf{u} \leftrightarrow \mathbf{v}$  gives the same with  $\mathbf{u}$  on the right. Hence

$$(P_X \mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (P_X \mathbf{v}) \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

**Step 2 — transpose rule.** Previous slide:  $(P_X \mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (P_X^T \mathbf{v})$ .

**Step 3 — conclude symmetry.** Steps 1 and 2 give  $\mathbf{u} \cdot (P_X^T \mathbf{v}) = \mathbf{u} \cdot (P_X \mathbf{v})$  for all  $\mathbf{u}, \mathbf{v}$ . Fix  $\mathbf{v} = \mathbf{e}_j$  and  $\mathbf{u} = \mathbf{e}_i$ : the  $j$ th column of  $P_X^T$  equals the  $j$ th column of  $P_X$  for every  $j$ . So  $P_X^T = P_X$ .

Direct check:  $(P_X)^T = X((X^T X)^{-1})^T X^T = P_X$ , using  $(X^T X)^{-1} = ((X^T X)^T)^{-1} = (X^T X)^{-1}$ .

## A.5 Orthogonal and orthonormal matrices

for interest only

**Orthonormal columns.** A square matrix  $Q \in \mathbb{R}^{n \times n}$  is **orthogonal** if

$$Q^T Q = I \quad (\text{equivalently } Q Q^T = I).$$

Column  $i$  is  $\mathbf{q}_i$ ; then  $\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$  (orthonormal).

**Geometric meaning.**  $Q$  is a **rigid motion** of  $\mathbb{R}^n$  (rotation and/or reflection):

$$\|Q\mathbf{x}\| = \|\mathbf{x}\|, \quad (Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T \mathbf{y}.$$

Lengths and angles are preserved;  $Q^{-1} = Q^T$ .

**Terminology.** “Orthonormal matrix” usually means the same as orthogonal (unit-length orthonormal columns).

## A.5 Symmetric matrices — eigenvalues are real

for interest only

**Claim.** If  $A = A^T$  (real symmetric), every eigenvalue  $\lambda$  is **real**.

**Proof.** Let  $A\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$  (allow complex  $\mathbf{v}$ ). Multiply on the left by  $\mathbf{v}^H$  (conjugate transpose):

$$\mathbf{v}^H A \mathbf{v} = \lambda \mathbf{v}^H \mathbf{v}.$$

Since  $A$  is real and symmetric,  $(\mathbf{v}^H A \mathbf{v})^* = \mathbf{v}^H A \mathbf{v}$ , so the left side is **real**. Also  $\mathbf{v}^H \mathbf{v} = \|\mathbf{v}\|^2 > 0$ . Hence  $\lambda = (\mathbf{v}^H A \mathbf{v})/(\mathbf{v}^H \mathbf{v}) \in \mathbb{R}$ .

**Not all symmetric eigenvalues are positive** — e.g.  $A = \text{diag}(1, -1)$ . Positive eigenvalues need extra assumptions (below).

## A.5 Positive semidefinite — eigenvalues $\geq 0$

for interest only

**Definition (PSD).** A symmetric matrix  $A$  is **positive semidefinite** if  $\mathbf{v}^T A \mathbf{v} \geq 0$  for all  $\mathbf{v} \in \mathbb{R}^n$ .

**Claim.** If  $A$  is PSD and  $A \mathbf{v}_i = \lambda_i \mathbf{v}_i$  with  $\mathbf{v}_i \neq \mathbf{0}$ , then  $\lambda_i \geq 0$ .

**Proof.** Along an eigenvector,

$$\lambda_i \mathbf{v}_i^T \mathbf{v}_i = \mathbf{v}_i^T A \mathbf{v}_i \geq 0 \quad \implies \quad \lambda_i \geq 0 \quad (\text{since } \mathbf{v}_i^T \mathbf{v}_i = \|\mathbf{v}_i\|^2 > 0).$$

**Examples.** Covariance matrices and  $A^T A$  are PSD, so their eigenvalues satisfy  $\lambda_i \geq 0$ . **Positive definite:**  $\mathbf{v}^T A \mathbf{v} > 0$  for all  $\mathbf{v} \neq \mathbf{0} \Rightarrow \lambda_i > 0$ .

## A.6 Equivalent linear systems

for interest only

**Equivalent systems.** Two linear systems (same unknowns) are **equivalent** if they have **exactly the same solutions**.

**Easier equivalent system.** Same solution set, but written in a form that is simpler to read — e.g. fewer redundant equations, leading variables isolated, zeros below each pivot. Row reduction replaces a system by an easier equivalent one **without changing who solves it**.

**Homogeneous case.** For  $M\mathbf{x} = \mathbf{0}$ , every equivalent system has the same **kernel** (null space)

$$\ker M = \{\mathbf{x} : M\mathbf{x} = \mathbf{0}\}.$$

Finding eigenvectors for  $\lambda$  is the same problem with  $M = A - \lambda I$ .

## A.6 Elementary row operations

for interest only

**Allowed operations** on a matrix (or augmented matrix  $[M \mid \mathbf{b}]$ ):

1. **Swap** two rows.
2. **Scale** one row by a **nonzero** number.
3. **Add** a multiple of one row to another row.

Each operation produces an **equivalent system** (same solution set).

**Why solutions are preserved (idea).** Swapping or scaling an equation does not change its truth set. If row  $i$  and row  $j$  are both satisfied, so is  $(\text{row } j) + c (\text{row } i)$ ; conversely you can undo the combination using the other allowed ops.

**RREF.** Keep applying these ops until the matrix is in **reduced row echelon form**: leading 1 in each pivot row; only zero above/below each leading 1; each leading 1 lies to the right of the one above.

## A.6 Pivot variables and free variables

for interest only

Once a matrix is in **RREF**, each column is either a **pivot column** or a **free column**.

**Pivot column / pivot variable.** A column with a **leading** 1; the matching unknown is a **pivot variable**. Its RREF row expresses the pivot variable as an **affine function of the free variables**.

**Free column / free variable.** A column *without* a leading 1; the matching unknown is a **free variable**. You may set each free variable to **any** value; nothing in RREF forces it.

**Key step.** Read each pivot row and **write each pivot variable as a function of the free variables**:

$$x_{\text{pivot}} = (\text{expression involving only free variables}).$$

Once the free variables are chosen, every pivot variable is determined; the whole solution is fixed.



## A.6 Example: row reduction (non-trivial)

for interest only

**Problem.** Find all  $\mathbf{x} \in \mathbb{R}^4$  with  $M\mathbf{x} = \mathbf{0}$  for

$$M = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 2 & 0 & 4 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{pmatrix}.$$

(Coupled equations; not block-diagonal.)

**Row ops.**  $R_2 \leftarrow R_2 - 2R_1$  eliminates the duplicate  $x_1$ -row;  $R_4 \leftarrow R_4 - 2R_3$  eliminates the duplicate  $x_2$ -row:

$$\longrightarrow \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{swap } R_2, R_3} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{RREF}).$$

## A.6 Example: pivot variables as functions of free variables

for interest only

From the RREF above: pivot columns 1, 2  $\Rightarrow$  pivot variables  $x_1, x_2$ ; free columns 3, 4  $\Rightarrow$  free variables  $x_3, x_4$ .

Equivalent system:

$$x_1 + 2x_3 + x_4 = 0, \quad x_2 + x_3 = 0.$$

Solve each pivot row for its pivot variable:

$$x_1 = -2x_3 - x_4, \quad x_2 = -x_3, \quad x_3, x_4 \in \mathbb{R} \text{ free.}$$

Every solution is fixed once  $(x_3, x_4)$  is chosen.

## A.6 Basis of the kernel from free variables

for interest only

**General solution.** Substitute the free variables into  $\mathbf{x} = (x_1, x_2, x_3, x_4)^\top$ :

$$\mathbf{x} = x_3 \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

So  $\ker M = \text{span}\{\mathbf{b}_1, \mathbf{b}_2\}$  with  $\dim \ker M = 2$  (two free variables).

**Recipe (basis of  $\ker M$ ).** For each free variable in turn:

1. set that free variable = 1;
2. set all other free variables = 0;
3. compute pivot variables from the RREF rows.

Here:  $(x_3, x_4) = (1, 0) \Rightarrow \mathbf{b}_1 = (-2, -1, 1, 0)^\top$ ;  $(x_3, x_4) = (0, 1) \Rightarrow \mathbf{b}_2 = (-1, 0, 0, 1)^\top$ .

Check:  $M\mathbf{b}_1 = \mathbf{0}$ ,  $M\mathbf{b}_2 = \mathbf{0}$ .

## A.7 When eigenvalues are not real

for interest only

The characteristic equation  $\det(A - \lambda I) = 0$  may have **no real roots**. Then  $A$  has **no real eigenvectors** (no real direction kept by scaling alone).

**Example:** 90° rotation (from earlier slides),  $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ :

$$\det(A - \lambda I) = \lambda^2 + 1 = 0 \quad \Rightarrow \quad \lambda = \pm i \text{ (not real).}$$

Every nonzero vector is rotated off its own line, so no  $\mathbf{v} \in \mathbb{R}^2 \setminus \{\mathbf{0}\}$  satisfies  $A\mathbf{v} = \lambda\mathbf{v}$ .

**Over  $\mathbb{C}$ .** Allow complex  $\lambda$  and  $\mathbf{v}$ ; the same recipe (characteristic polynomial, then  $(A - \lambda I)\mathbf{v} = \mathbf{0}$ ) works. For **real**  $A$ , non-real eigenvalues come in **conjugate pairs**  $\lambda, \bar{\lambda}$ .

Existence over  $\mathbb{C}$ : [appendix \(for interest only\)](#). When a real eigenbasis fails, see A.8 for QR and SVD (brief).

## A.8 QR and SVD (brief)

for interest only

**Same spirit as eigen-decomposition:** rewrite  $A$  so the action is easy to read.

- **QR** ( $A = QR$ ):  $Q$  has orthonormal columns,  $R$  is upper triangular. Used for stable least squares and picking independent columns ([appendix](#)).
- **SVD** ( $A = U\Sigma V^T$ ): always exists;  $U, V$  orthogonal,  $\Sigma$  diagonal (singular values). Scales along orthogonal directions — works even when real eigenvectors do not exist (e.g. rotation).

More detail on main slide “Decompositions — QR, eigen, SVD” (for interest only). Existence: [appendix](#), [appendix](#) (**for interest only**).

## A.9 Existence of eigenvalues and eigenvectors (over $\mathbb{C}$ )

for interest only

**Claim.** Every  $A \in \mathbb{C}^{n \times n}$  has at least one eigenpair  $(\lambda, \mathbf{v})$  with  $\mathbf{v} \neq \mathbf{0}$ .

**Step 1 (characteristic polynomial).**  $p_A(\lambda) = \det(A - \lambda I)$  is a polynomial in  $\lambda$  of degree  $n$ .

**Step 2 (eigenvalue exists).** By the **fundamental theorem of algebra**,  $p_A(\lambda) = 0$  has  $n$  roots in  $\mathbb{C}$  (counting multiplicity). Pick one root  $\lambda$ .

**Step 3 (eigenvector exists).**  $p_A(\lambda) = 0$  means  $\det(A - \lambda I) = 0$ , so  $A - \lambda I$  is **singular**. Hence  $(A - \lambda I)\mathbf{v} = \mathbf{0}$  has a nontrivial solution  $\mathbf{v} \neq \mathbf{0}$ .

Real  $A$ : non-real roots come in conjugate pairs  $\lambda, \bar{\lambda}$ .

## A.10 Existence of QR — what the decomposition means

for interest only

**Claim.** Every  $A \in \mathbb{R}^{m \times n}$  can be written as  $A = QR$ :  $Q$  has **orthonormal columns**,  $R$  is **upper triangular**.

**Setup.** Write  $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \cdots]$  — a **series of column vectors** 1, 2, 3, ... They may overlap (not perpendicular, different lengths).

**Goal (Gram–Schmidt).** Build new **clean directions**  $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots$  (perpendicular, unit length) so that  
vector  $j$  is built only from  $\mathbf{q}_1, \dots, \mathbf{q}_j$ .

The numbers  $r_{ij}$  in  $R$  record **how much** of each vector goes along each direction.

## A.10 Existence of QR — vectors 1 and 2

for interest only

**Vector 1.** Take the first column, normalize it — that is the first clean direction:

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|}.$$

Vector 1 lies entirely along  $\mathbf{q}_1$ .

**Vector 2.** Project  $\mathbf{a}_2$  onto  $\mathbf{q}_1$ , subtract, and keep the **residual** as the new direction:

$$\mathbf{a}_2 = \underbrace{(\mathbf{q}_1^T \mathbf{a}_2) \mathbf{q}_1}_{\text{part along } \mathbf{q}_1} + \underbrace{(\mathbf{a}_2 - (\mathbf{q}_1^T \mathbf{a}_2) \mathbf{q}_1)}_{\perp \mathbf{q}_1}.$$

Normalize the residual to get  $\mathbf{q}_2$ . **Key:** vector 2 uses **only**  $\mathbf{q}_1$  and  $\mathbf{q}_2$ :  $\mathbf{a}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$ .



## A.10 Existence of QR — vectors 3, 4, ...

for interest only

**Vector 3.** Project  $\mathbf{a}_3$  onto  $\mathbf{q}_1$  and  $\mathbf{q}_2$ , subtract both overlaps. Normalize what's left  $\Rightarrow \mathbf{q}_3$ . Vector 3 uses **only**  $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ .

**Vector 4, ...** Same pattern: for vector  $k$ , subtract its projections onto all previous clean directions, normalize the residual  $\Rightarrow \mathbf{q}_k$ .

**General step.**

$$\mathbf{a}_k = \sum_{j=1}^{k-1} (\mathbf{q}_j^T \mathbf{a}_k) \mathbf{q}_j + (\text{residual } \perp \text{ all previous } \mathbf{q}_j).$$

So  $\mathbf{a}_k = r_{1k}\mathbf{q}_1 + \cdots + r_{kk}\mathbf{q}_k$  — no  $\mathbf{q}_\ell$  with  $\ell > k$  needed.

## A.10 Existence of QR — reading $R$ and $A = QR$

for interest only

**What is  $r_{ij}$ ?** From  $\mathbf{a}_j = \sum_i \mathbf{q}_i r_{ij}$ : entry  $r_{ij}$  is **how much of vector  $j$  lies along direction  $i$ :**

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j.$$

**Why upper triangular?** Vector  $j$  is built from  $\mathbf{q}_1, \dots, \mathbf{q}_j$  only, so  $r_{ij} = 0$  for  $i > j$ . Diagonal  $r_{jj} = \text{length of the residual}$  (the genuinely new part).

**Existence.** Run the process on all columns; stack  $\mathbf{q}_j$  into  $Q$  and  $r_{ij}$  into  $R$ . Then  $A = QR$ .

## A.11 Existence of SVD — intuition

for interest only

**Dimensions.**  $A \in \mathbb{R}^{m \times n}$  maps  $\mathbb{R}^n \rightarrow \mathbb{R}^m$ :  $\mathbf{x} \in \mathbb{R}^n \Rightarrow A\mathbf{x} \in \mathbb{R}^m$ .

**Why SVD?** Eigenvectors need  $A\mathbf{v} = \lambda\mathbf{v}$  (same direction in and out) — fine for square  $A$  with a full eigenbasis. Fails for rotation (no real eigenvectors) and when  $m \neq n$  (input and output live in different spaces).

**Idea.** Find **unit** input directions  $\mathbf{v}_i$  ( $\|\mathbf{v}_i\| = 1$ ) and unit output directions  $\mathbf{u}_i$  ( $\|\mathbf{u}_i\| = 1$ ) with

$$A\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \sigma_i \geq 0.$$

Input direction  $\mathbf{v}_i$  maps to output direction  $\mathbf{u}_i$ , stretched by  $\sigma_i$  (no mixing).

**Picture.** Unit sphere in  $\mathbb{R}^n \rightarrow$  ellipsoid in  $\mathbb{R}^m$ .  $\mathbf{v}_i$  = input axes;  $\mathbf{u}_i$  = where they land;  $\sigma_i$  = axis lengths. Factorization:  $A = U\Sigma V^T$  = rotate input ( $V$ ), scale ( $\Sigma$ ), rotate output ( $U$ ).

## A.11 Existence of SVD — derivation (Step 1)

for interest only

**Step 1 — input directions ( $\mathbb{R}^n$ ).** Diagonalize  $A^T A$  ( $n \times n$ ): orthonormal eigenvectors  $\mathbf{v}_i$  with  $\|\mathbf{v}_i\| = 1$ , eigenvalues  $\lambda_i \geq 0$ . Set  $\sigma_i = \sqrt{\lambda_i} = \|\mathbf{A}\mathbf{v}_i\|$  (stretch along unit input  $\mathbf{v}_i$ ).

**Remark.**  $\lambda_i$  is real (appendix) and  $\lambda_i \geq 0$  (appendix) because  $A^T A$  is symmetric PSD. So  $\sigma_i = \sqrt{\lambda_i}$  is real and  $\geq 0$ ; it is **not** an eigenvalue of  $A$ .

## A.11 Existence of SVD — derivation (Steps 2–3)

for interest only

**Step 2 — output directions ( $\mathbb{R}^m$ ).** Goal:  $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$  with  $\|\mathbf{u}_i\| = 1$ . From Step 1:  $\mathbf{v}_i$  is **normalized** ( $\|\mathbf{v}_i\| = 1$ ) and  $\sigma_i = \|A\mathbf{v}_i\|$ .

Apply  $A$  to the unit input  $\mathbf{v}_i$ : the vector  $A\mathbf{v}_i \in \mathbb{R}^m$  points the right way but has length  $\sigma_i$ . Normalize it:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i \quad \implies \quad A\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \|\mathbf{u}_i\| = 1.$$

**Step 3 — assemble.**  $V = [\mathbf{v}_1 \cdots]$ ,  $U = [\mathbf{u}_1 \cdots]$ ,  $\Sigma = \text{diag}(\sigma_1, \dots) \Rightarrow A = U\Sigma V^\top$ .